

G11-01-LOCAL-ENERGY-MONOTONICITY-KILL-v1: Does $T^* < \infty$ force monotone M incompatible with C_{PROVEN} ?

ZFL-CX

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$G7 = FROZEN$, $G8 = CLOSED/BLOCKED$, $G9 BLOCKED$, $G10 CLOSED$, $G11-FRONTIER = DON$

G11-01 = local energy/monotonicity, NOT parallel with other 2

$$C_{PROVEN} = \{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}\}$$

1 Question

$$T^* < \infty \xrightarrow{C_{PROVEN}} concentration \xrightarrow{?} M(t, r) \text{ monotone} \xrightarrow{?} \perp$$

M must be more than norm disguised: $M \lesssim \|u\|_X \Rightarrow KILL (G8 \text{ reduction})$.

2 4 requirements

1. Exact definition of M : Candidate must be local/scale-adapted: e.g. $M(r, t) = r^{-1} \int_{B_r(x^*)} |u|^2 dx$, or $M(r) = r^{-1} \int_{B_r} |\omega|^2$, or flux $\int_{B_r} (u \cdot \nabla) u \cdot \phi$, or defect measure. Must be defined from suitable weak solution without new hypothesis.

2. Exact evolution/monotonicity law: Need identity/inequality $\partial_t M \leq 0$ or ≥ 0 or $\frac{d}{dr} M \geq -error$, derived from NS + local energy inequality. No assumed monotonicity. Known: for NS there is NO known monotone quantity like Struwe for harmonic map. CKN scaled energy $r^{-1} \int_{B_r} |u|^2$ is bounded but not monotone. So need proof.

3. Quantitative entry of C_F^{conc} : How does $\|u(t_n)\|_{L^3(B_{r_n})} \geq \varepsilon_0$ force $M(t_n, r_n)$ to have sign/behavior incompatible with global bound? C_F^{conc} is lower L^3 concentration. M typically is L^2 scaled: by Hölder $r^{-1} \|u\|_{L^2(B_r)}^2 \geq cr \|u\|_{L^3(B_r)}^2$? Check: $\|u\|_{L^3(B_r)} \leq Cr^{-1/2} \|u\|_{L^2}^2$ Actually scaling: $\|u\|_{L^3(B_r)} \leq \|u\|_{L^2(B_r)}^2$ Not give lower for M . So C_F^{conc} gives lower bound for M ? Estimate: $\|u\|_{L^3(B_r)} \leq Cr^{-1/2} (\int_{B_r} |u|^2)^{1/2}$? Then $M = r^{-1} \int |u|^2 \geq cr \|u\|_{L^3}^2 \geq cr \varepsilon_0^2 \rightarrow 0$. So lower bound degenerates to 0, no contradiction. Opposite direction again.

4. Contradiction location: Where does $T^* < \infty$ + monotonicity of M contradict C_E^{weak} or global? Must produce $\perp$, not just boundedness.

3 Check against $\mathcal{C}_{PROVEN}$

- With only suitable structure, local energy inequality gives:

$$\int \phi |u(t)|^2 + 2 \int \int \phi |\nabla u|^2 \leq \int \int |u|^2 (\partial_t \phi + \Delta \phi) + \int \int (|u|^2 + 2p) u \cdot \nabla \phi.$$

No monotone M emerges. Right-hand side contains cubic terms with no sign.

- Any attempt to define $M = r^{-1} \int_{B_r} |u|^2$ yields after scaling: $M(r_n, t_n) = r_n^{-1} \int_{B_{r_n}} |u(t_n)|^2$.

From C_F^{conc} : $\varepsilon_0 \leq \|u\|_{L^3(B_{r_n})} \leq C r_n^{-1/2} M^{1/2}$ so $M \geq c r_n \varepsilon_0^2 \rightarrow 0$. So lower bound not incompatible, tends to 0, no blow-up of M .

- To get monotone, would need smallness of pressure term $\int p u \cdot \nabla \phi$, which requires $L^{3/2}$ control not in $\mathcal{C}_{PROVEN}$. Without it, sign not controlled.

- $C_I^{freq} N_c \rightarrow \infty$ and C_J^{neg} loss of direction make error terms larger, not smaller.

Thus: No M satisfying 1-4 derivable from $\mathcal{C}_{PROVEN}$ without new hypothesis (smallness, Type I, $L_t^\infty L^3$).

$$M \lesssim \|u\|_X \text{ or } M \geq c r_n \varepsilon_0^2 \rightarrow 0 \Rightarrow \text{no } \perp$$

NS has no known local monotone quantity with sign from suitable alone

4 Kill verdict

G11-01 Gate 1-4 = BLOCKED with $\mathcal{C}_{PROVEN}$

M definition would require extra bound to control pressure flux and produce monotonicity; C_F^{conc} enters only as vanishing lower bound for M , not as forced growth.

Therefore:

G11-01 = CLOSED/BLOCKED

G11 family local-energy/monotonicity CLOSED with current C_i

This does NOT kill other 2 families (vorticity topology, alt ancient rigidity), but with current $\mathcal{C}_{PROVEN}$ local-energy route is blocked clean, no parallel opening.

G7 FROZEN $\rightarrow$ G8 CLOSED $\rightarrow$ G9 BLOCKED $\rightarrow$ G10 CLOSED $\rightarrow$ G11-01 CLOSED

STOP = PROGRESO, map intact